

Chapter 4.

4.1.

$$\begin{bmatrix} 1 & -1 \\ -2 & 5 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} 3 & -2 \\ -2 & 1 \end{bmatrix}$$

4.2.

$$\begin{bmatrix} 2 & 1 \\ 1 & -1 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 3 & 2 \\ 3 & 1 & 0 & 5 \end{bmatrix}$$

4.3.

$$\begin{bmatrix} 2 & -3 & 5 \\ -1 & 2 & -3 \\ 7 & 2 & 6 \end{bmatrix} \begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{bmatrix} = \begin{bmatrix} -1 & 2 & 1 \\ 0 & -5 & -3 \\ 2 & 0 & 10 \end{bmatrix}$$

4.4.

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & -2 & 1 \\ -1 & 1 & 3 \end{bmatrix}$$

4.5.

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$$

4.6.

(a) $L = \begin{bmatrix} 1 & & \\ 2 & 1 & \\ 1 & -1 & 1 \end{bmatrix}$ $D = \begin{bmatrix} 2 & & \\ & -1 & \\ & & 6 \end{bmatrix}$ $U = \begin{bmatrix} 1 & 2 & \frac{1}{2} \\ & 1 & -1 \\ & & 1 \end{bmatrix}$

(b) $L = \begin{bmatrix} 1 & & \\ -\frac{2}{3} & 1 & \\ & 3 & 1 \end{bmatrix}$ $D = \begin{bmatrix} -1 & & \\ & 2 & \\ & & -20 \end{bmatrix}$ $U = \begin{bmatrix} 1 & & -3 \\ & 1 & \frac{5}{2} \\ & & 1 \end{bmatrix}$

(c) $L = \begin{bmatrix} 1 & & \\ 0 & 1 & \\ -3 & -5 & -\frac{22}{9} \end{bmatrix}$ $D = \begin{bmatrix} 1 & & \\ & -1 & \\ & & -9 \\ & & & -\frac{5}{9} \end{bmatrix}$ $U = \begin{bmatrix} 1 & 2 & 3 & 4 \\ & 1 & -2 & 4 \\ & & 1 & -\frac{1}{9} \\ & & & 1 \end{bmatrix}$

47. (i) $LZ = b$ to find Z
 (ii) $DY = Z$ to find Y
 (iii) $UX = Y$ to find X .

48. (a) $X = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$ (b) $X = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}$

49. Let $A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}$ $B = \begin{bmatrix} b_1 & & & \\ & b_2 & & \\ & & \ddots & \\ & & & b_n \end{bmatrix}$

Then $AB = \begin{bmatrix} b_1 a_{11} & b_2 a_{12} & \dots & b_n a_{1n} \\ b_1 a_{21} & b_2 a_{22} & \dots & b_n a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_1 a_{n1} & b_2 a_{n2} & \dots & b_n a_{nn} \end{bmatrix}$

Chapter 5

5.1. Interchanges of columns

5.2. $X = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

5.3. $X = \begin{bmatrix} -7 \\ 6 \\ -2 \\ 0 \end{bmatrix}$

5.4. (a) $L = \begin{bmatrix} 1 & 0 \\ \frac{1}{3} & 1 \end{bmatrix}$ $U = \begin{bmatrix} 3 & 4 \\ 0 & \frac{2}{3} \end{bmatrix}$ $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

(b) $L = \begin{bmatrix} 1 & & \\ -\frac{1}{2} & 1 & \\ \frac{1}{2} & -\frac{1}{3} & 1 \end{bmatrix}$ $U = \begin{bmatrix} -2 & 4 & 8 \\ 0 & 3 & 5 \\ 0 & 0 & -3 \end{bmatrix}$ $P = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

$$5.5. \quad (a) X = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

$$(b) X = \begin{bmatrix} 3/5 \\ 0 \\ 7/5 \end{bmatrix}$$

$$5.6. \quad \sim \frac{2}{3}n^3$$

$$5.7. \quad P_1 A P_2 = L U$$

where P_1 and P_2 are permutation matrices.

Chapter 6.

6.1. (a) Singular

(b) Nonsingular

(c) Singular

$$6.2. \quad (a) \begin{bmatrix} -5 & -4 \\ -1 & -1 \end{bmatrix}$$

$$(b) \begin{bmatrix} 2 & 1 & -1 \\ -1 & -1 & 1 \\ 1 & 2 & -1 \end{bmatrix}$$

$$(c) \begin{bmatrix} 1 & -3 & -2 \\ -4 & 1 & 1 \\ -2 & 1 & 0 \end{bmatrix}$$

$$6.3. \quad (a) X = \begin{bmatrix} -15 \\ -3 \end{bmatrix}$$

$$(b) = \begin{bmatrix} -3 \\ 2 \\ 0 \end{bmatrix}$$

$$(c) = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}$$

$$\begin{aligned} 6.4 \text{ Proof: } \begin{bmatrix} a & b \\ c & d \end{bmatrix} \cdot \left(\frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \right) &= \frac{1}{ad-bc} \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \\ &= \frac{1}{ad-bc} \begin{bmatrix} ad-bc & 0 \\ 0 & -bc+ad \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \end{aligned}$$

$$\text{So } \left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} \right)^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$6.5. \quad B = 0$$

$$6.6. \quad (A^T)^{-1} = (A^{-1})^T$$

$$6.7. \quad \left(\begin{bmatrix} 1 & 0 & 0 \\ -\ell_{21} & 1 & 0 \\ -\ell_{31} & 0 & 1 \end{bmatrix} \right)^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ \ell_{21} & 1 & 0 \\ \ell_{31} & 0 & 1 \end{bmatrix}$$

6.10. Let $A^{-1}B = C$, then

$$AC = A(A^{-1}B) = B.$$

Let $C = [C_1, C_2, \dots, C_m]$ where C_j is the j th-column of C .

$B = [b_1, b_2, \dots, b_m]$ where b_j is the j th-column of B .

So we can find C by solving $A[C_1, C_2, \dots, C_m] = [b_1, b_2, \dots, b_m]$

i.e, $AC_j = b_j$ for $j = 1, 2, \dots, m$.